



Transforming Network Engineers to Automation Practitioners with MCP

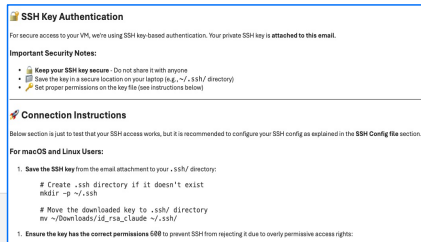


Josemi

Nilesch

Jessica

Access Credentials



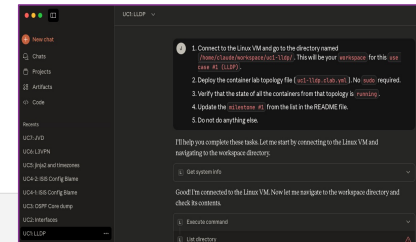
- ❑ Go through the email Josemi (Jose Miguel Izquierdo) sent titled “Autocon4 WS-A3 preparation” and add the correct SSH key and config

Request IP Address



- ❑ We whitelisted a block provided by NAF
- ❑ Find Josemi, Nilesch, or Jessica to add
- ❑ Check Connectivity

Other Applications



- ❑ Claude Desktop or GitHub Copilot recommended as AI Assistant
- ❑ VS Code recommended as IDE, specific plugins recommended, too

Follow the steps above to get ready for the workshop



Troubleshooting, FAQs, and more



Josemi

Nilesh

Jessica

PROBLEM: Cannot SSH to the VM

- Did you download the key?
- Did you add the SSH Config?
- Are you using an assigned IP address?

PROBLEM: Windows

RECOMMENDATIONS:

- WSL
- GitBash

PROBLEM: Can I use another IDE? LLM? AI Assistant?

Technically yes, we cannot stop you. We had someone use Zed.dev with Copilot in our internal run of the workshop. They handled all interoperability on their own.

PROBLEM: I cannot ping the VM

You will not be able to ping. Please test connectivity with the ssh commands in the email.

PROBLEM: Cannot open the repo in VSCode

WORKAROUND: Can use the repo on GitHub. The most important parts of the labs will work, but you will lack a few things.



Transforming Network Engineers into Automation Practitioners: AI-Powered Network Operations

Jose Miguel Izquierdo | Nilesch Simaria | Jessica Garrison

AutoCon4 Workshop | November 17, 2025

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Meet Your Workshop Proctors



```
proctor:  
  name: josemi izquierdo  
  lead: workshop  
  autocon_event_count: 4  
  juniper_years: 13  
  subjects:  
    - system integration  
    - llm/mcp  
    - automating on junos  
    - routing director
```



```
proctor:  
  name: Nilesh Simaria  
  lead: developer  
  autocon_event_count: 1  
  juniper_years: 17  
  subjects:  
    - c, go, Python  
    - llm/mcp  
    - automating on junos  
    - network monitoring
```



```
proctor:  
  name: jessica garrison  
  lead: slides  
  autocon_event_count: 3  
  juniper_years: 20  
  subjects:  
    - junos  
    - automating on junos  
    - product automation  
    - pyez/scripting  
    - ansible
```

Agenda

Pework from email

Intro & Demo

Seminar

Workshop

Agenda

Pework from email

Intro & Demo

Seminar

Workshop

- **If you did not receive an email, please talk to Josemi now**
- **SSH instructions from the email**
 - **Key**
 - **Config**
- **After we provide your IP address, you may test SSH access**

Agenda

Pework from email

Intro & Demo

Seminar

Workshop

- **Workshop Proctors**
- **AI Use Cases for Networking**
- **DEMO - Configuration Blame Analysis with ISIS**

Agenda

Pework from email

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Workshop

- **AI Evolution**
- **AI Systems [Predictive, Generative, & Agentic AI]**
- **Predictive AI**
- **Generative AI**
- **Agentic AI**
- **MCP (Model Context Protocol)**
- **Junos MCP server**

Agenda

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Seminar

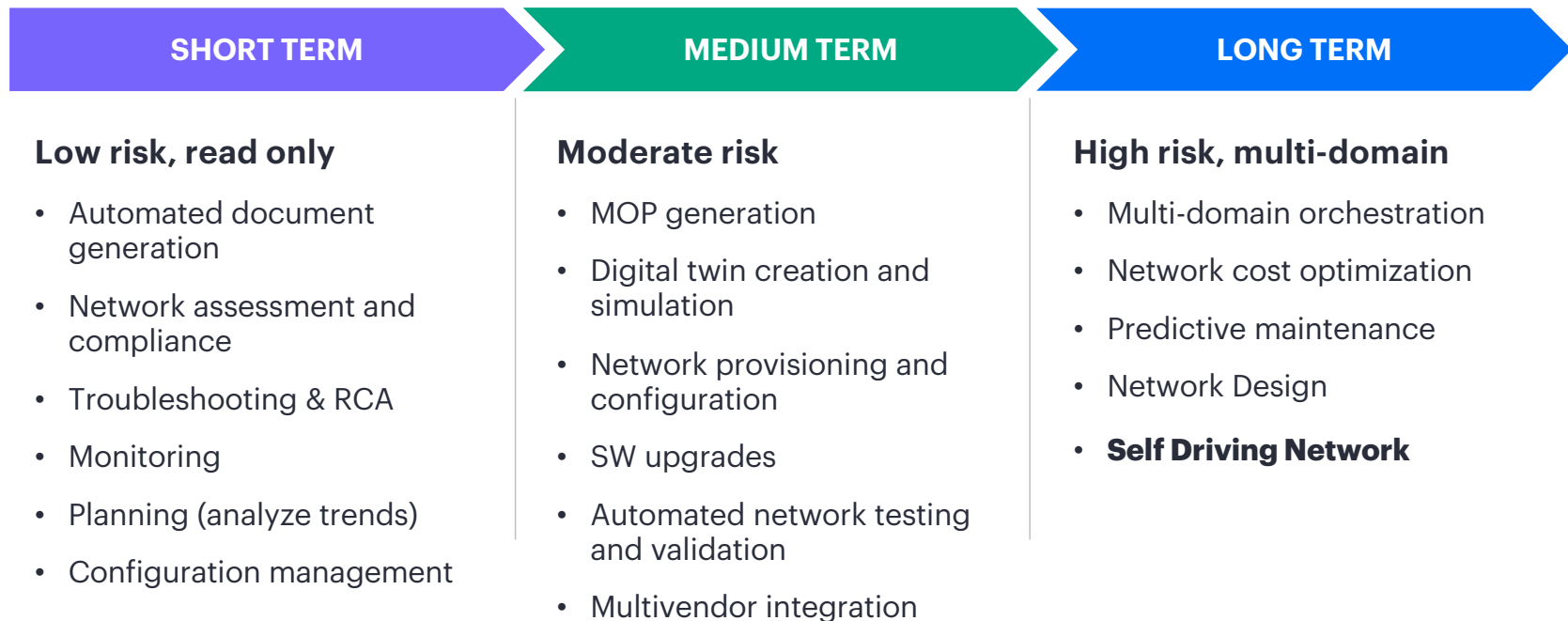
Workshop

- **Tips and Tricks**
- **Workshop VM**

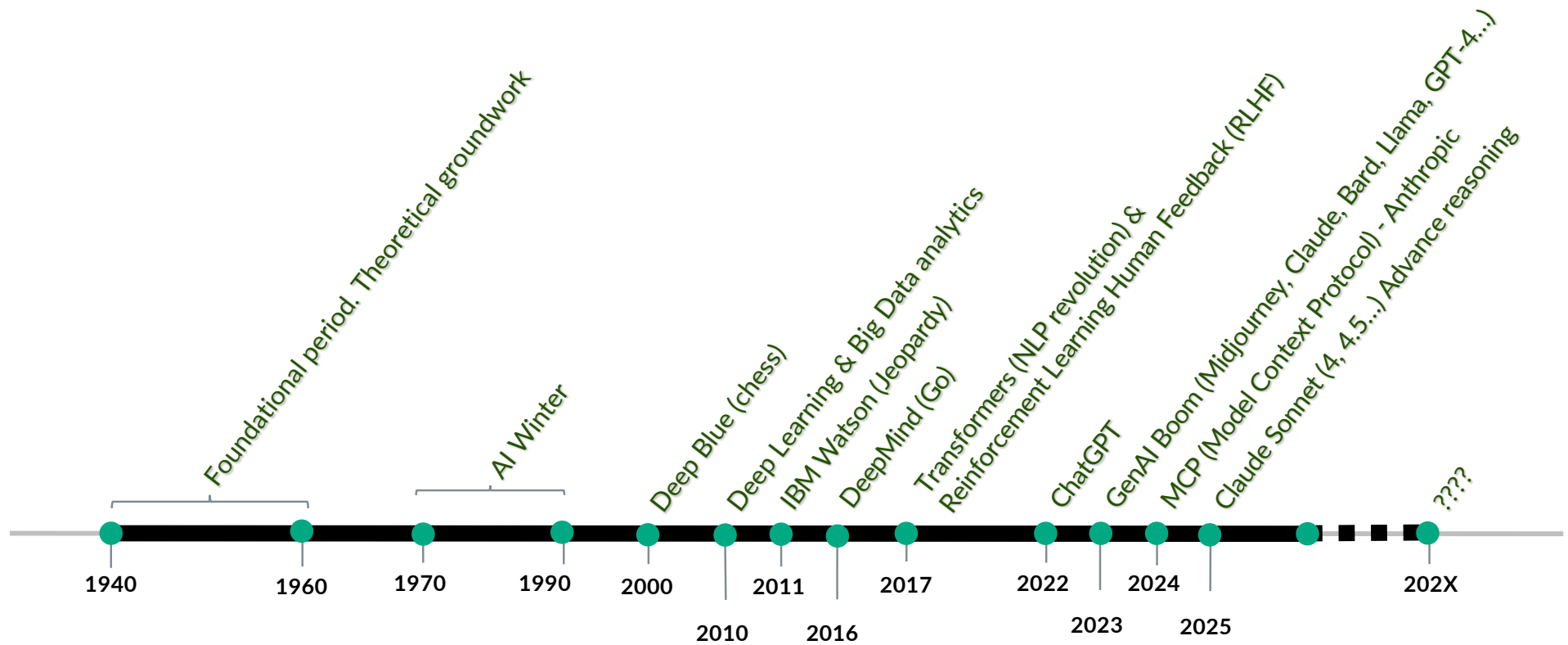
Use Cases:

- **Configuration Audit with Interface Descriptions**
- **Troubleshooting L3VPN**
- **Juniper Validated Design Deployment**
- **Design your own health check**

Agentic AI use-cases for the network



Timeline



#ac4-ws-a3

The different types of AI systems

Predictive



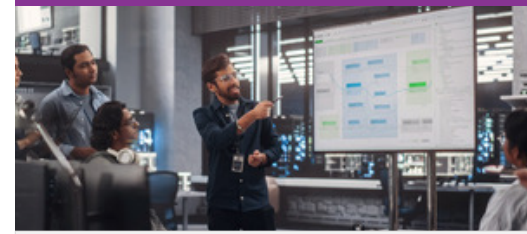
- ▣ Analyzes historical data to make future predictions
- ▣ Example: Bandwidth utilization in your network

Generative



- ❖ Creates original content based on patterns in training data
- ❖ Example: Configuration or code generation

Agentic



- ▶ Accomplish complex goals through planning and execution
- ▶ Example: Manage your end-to-end network

Prediction

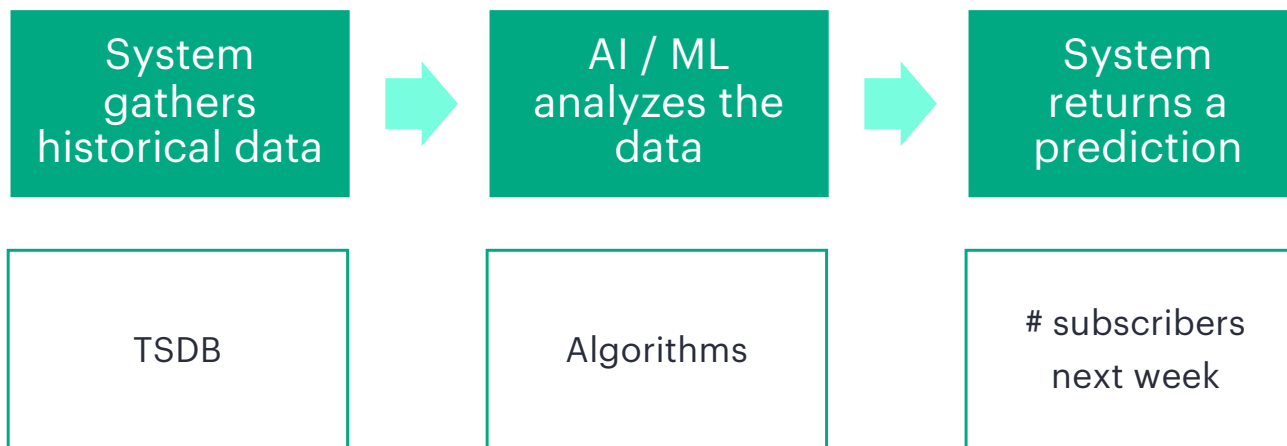
Creation

Action

Predictive AI

Learns from the past and forecasts outcomes

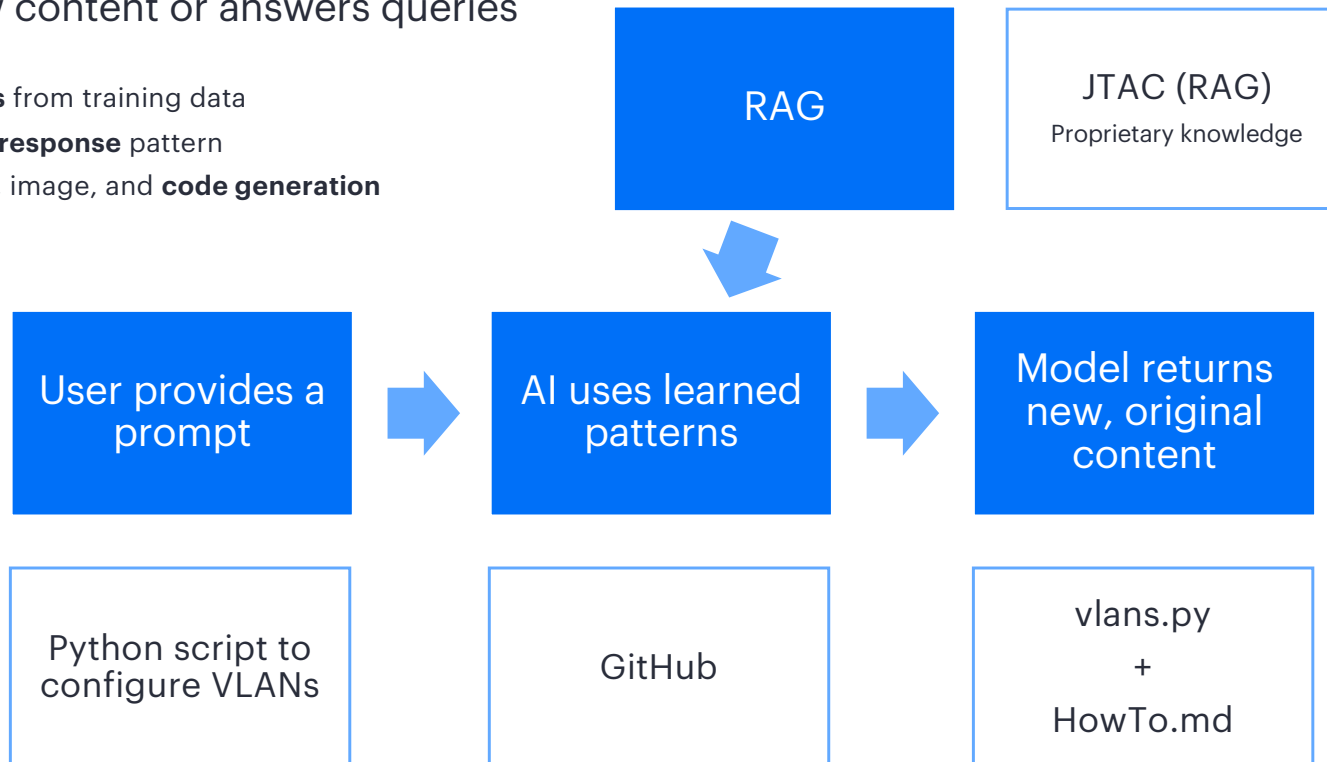
- **Analyses** historical and trending data, **identifies patterns/correlations**, and makes **data-driven predictions**
- Good for **risk assessment** and optimization
- Can use dynamic thresholds for **anomaly detection**
- **Examples:** Forecasting demand (retail or network bandwidth), predicting failures (automobiles or networking hardware), fraud detection



Generative AI

Creates new content or answers queries

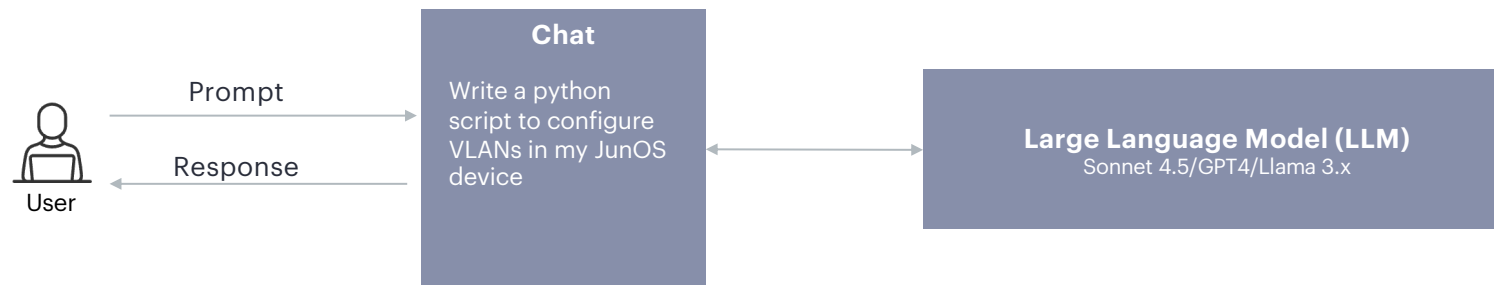
- Learns **patterns** from training data
- Single **prompt-response** pattern
- **Examples:** text, image, and **code generation**



Generative AI

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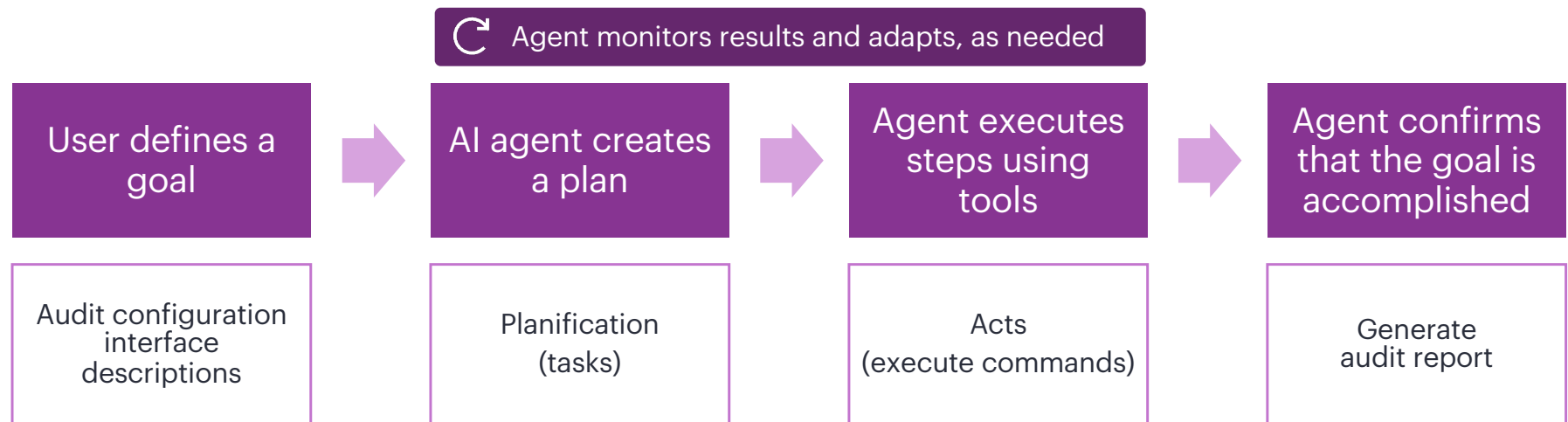
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- **Examples:** text, image, audio or **code generation**



Agentic AI

Autonomous, adaptive, and goal-oriented AI that can execute multi-step tasks

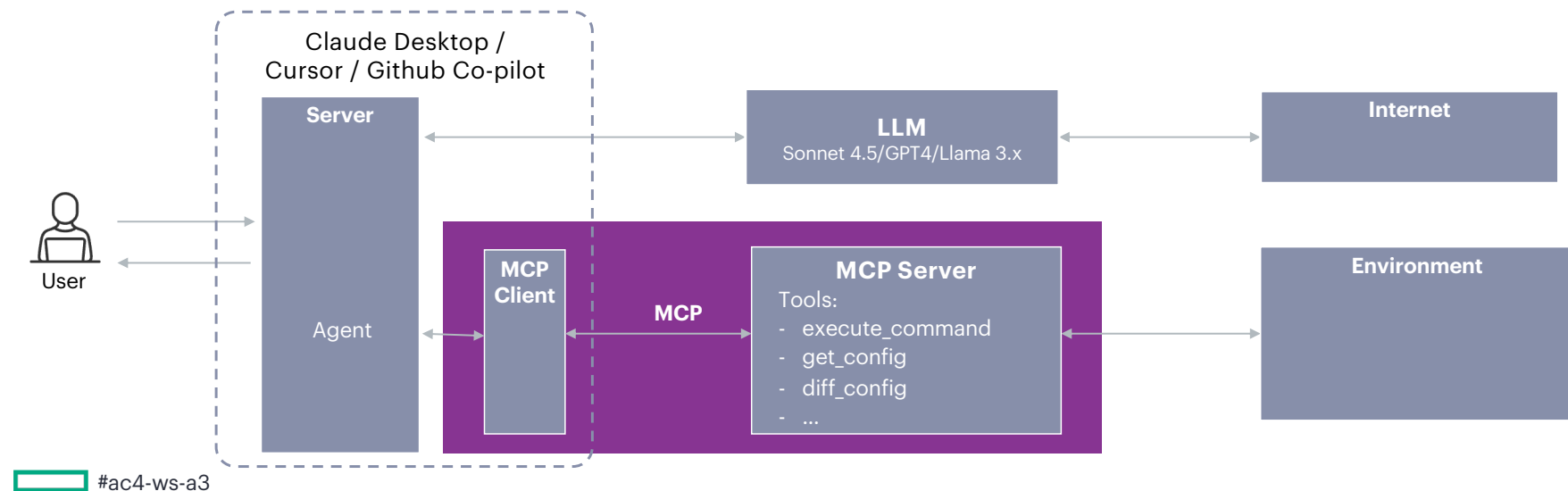
- Identifies the **user's high-level intent** to create and execute on a plan
- Selects the **right tool and APIs** to gather data and take action
- **Maintains context** across multiple interactions; learns from experience and iterations to improve over time
- **Examples:** Research features on products, develop code for network automation, create health check reports



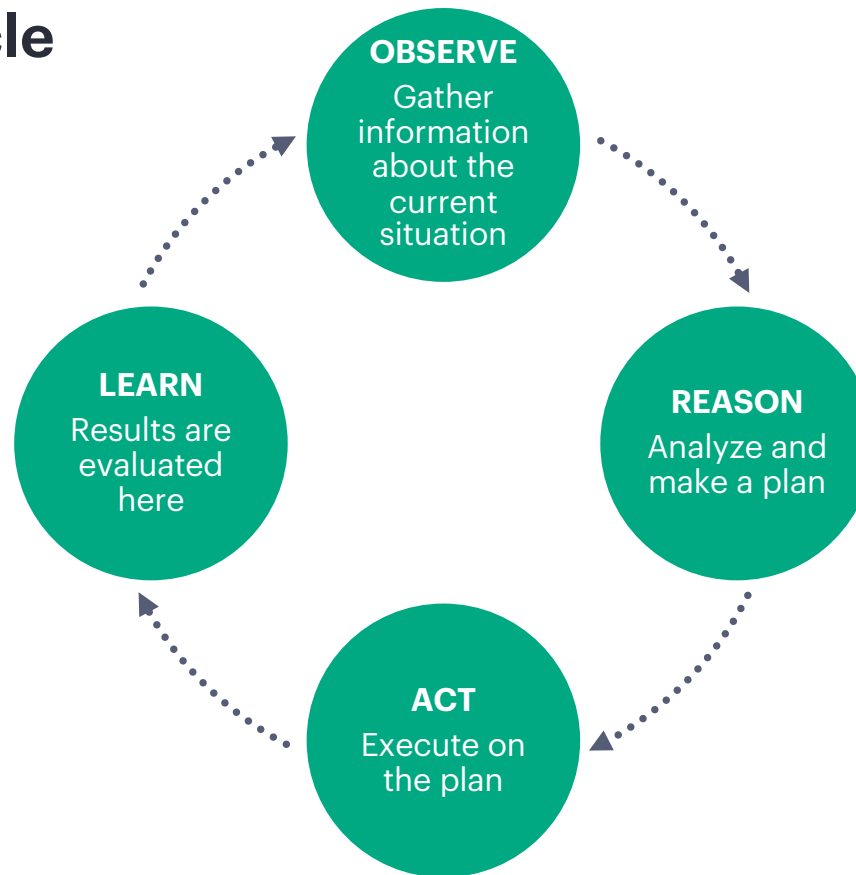
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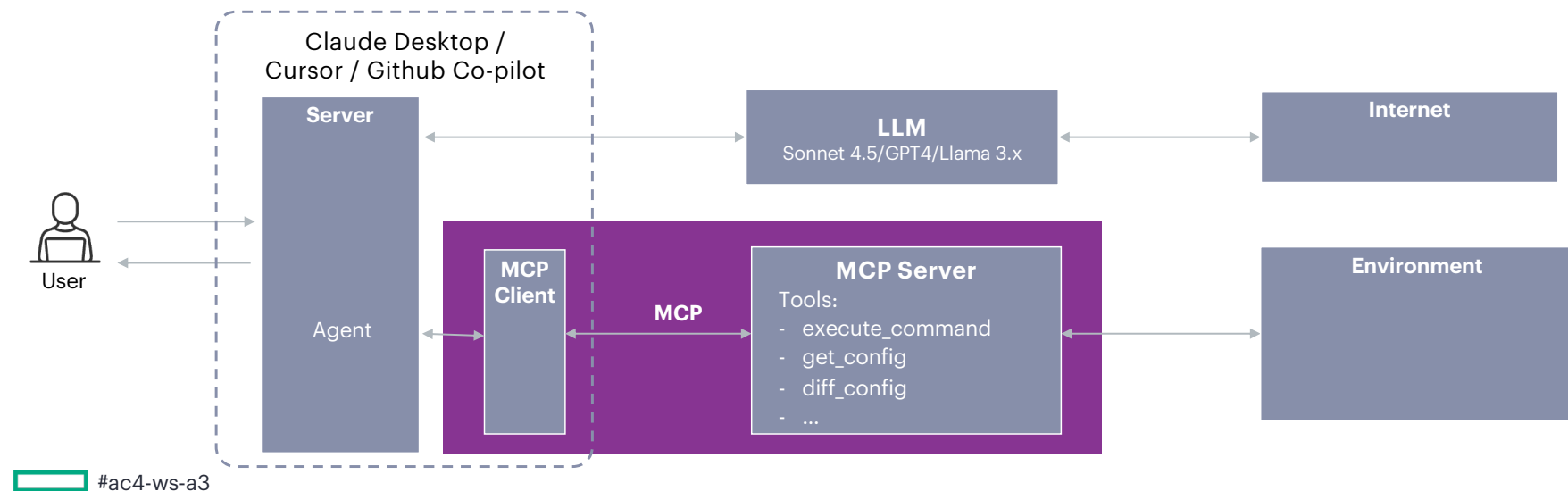
Agentic AI's continuous cycle



Agentic AI

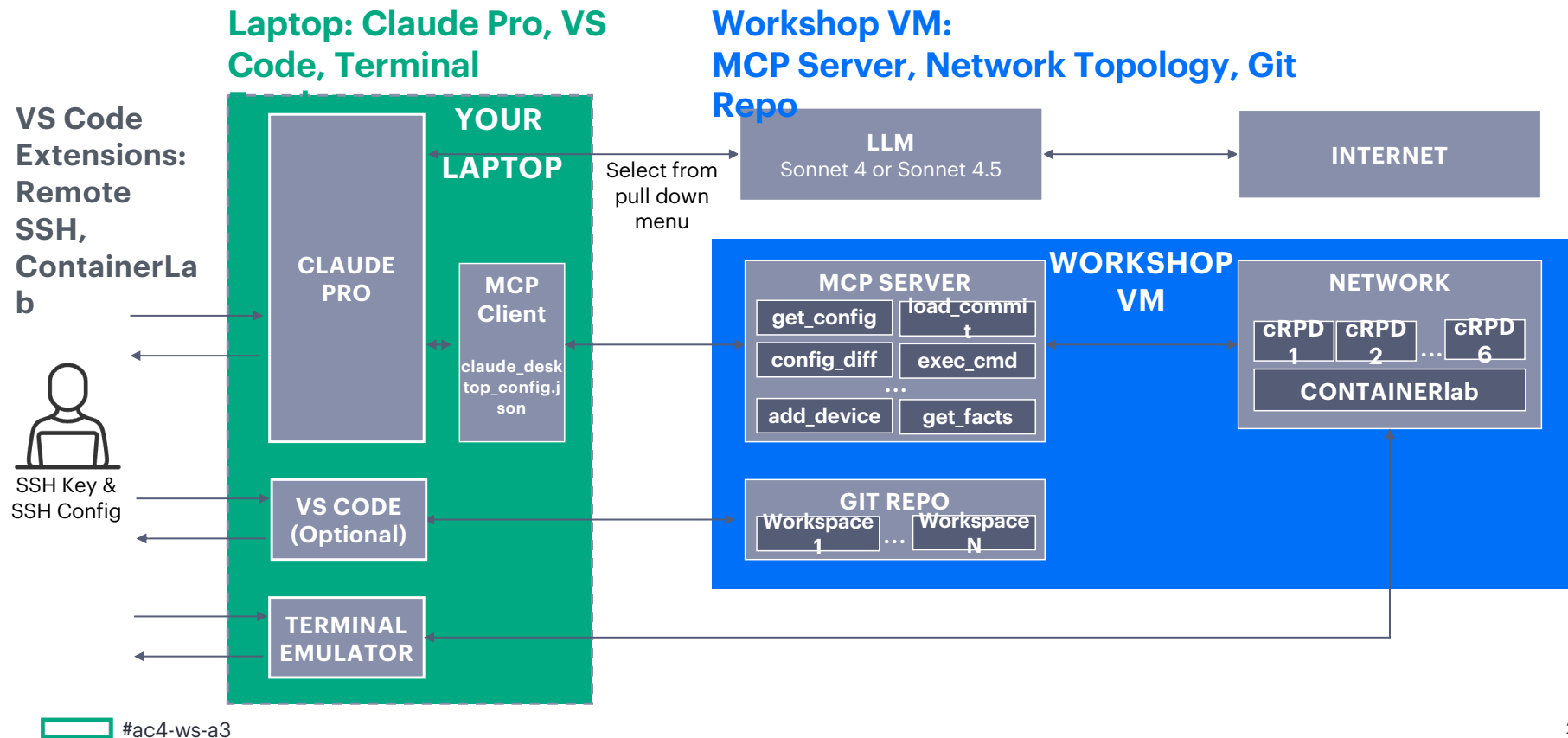
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#ac4-ws-a3

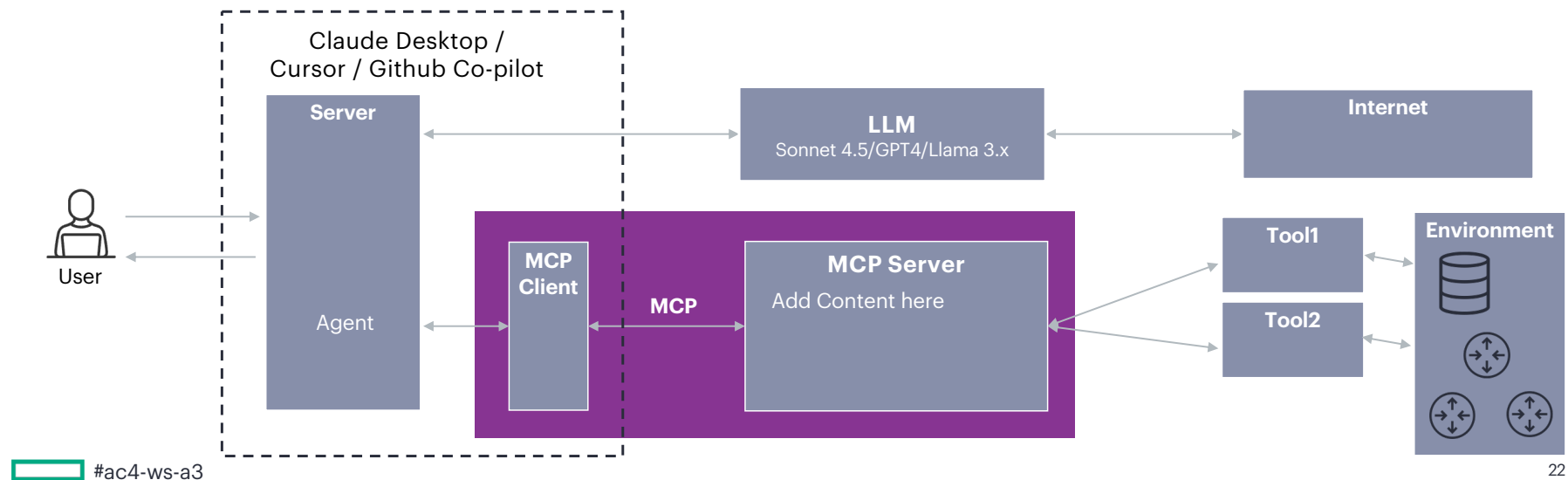
What does our lab topology look like?



Model Context Protocol (MCP)

Adapter that lets AI connect to the user's world to accomplish tasks

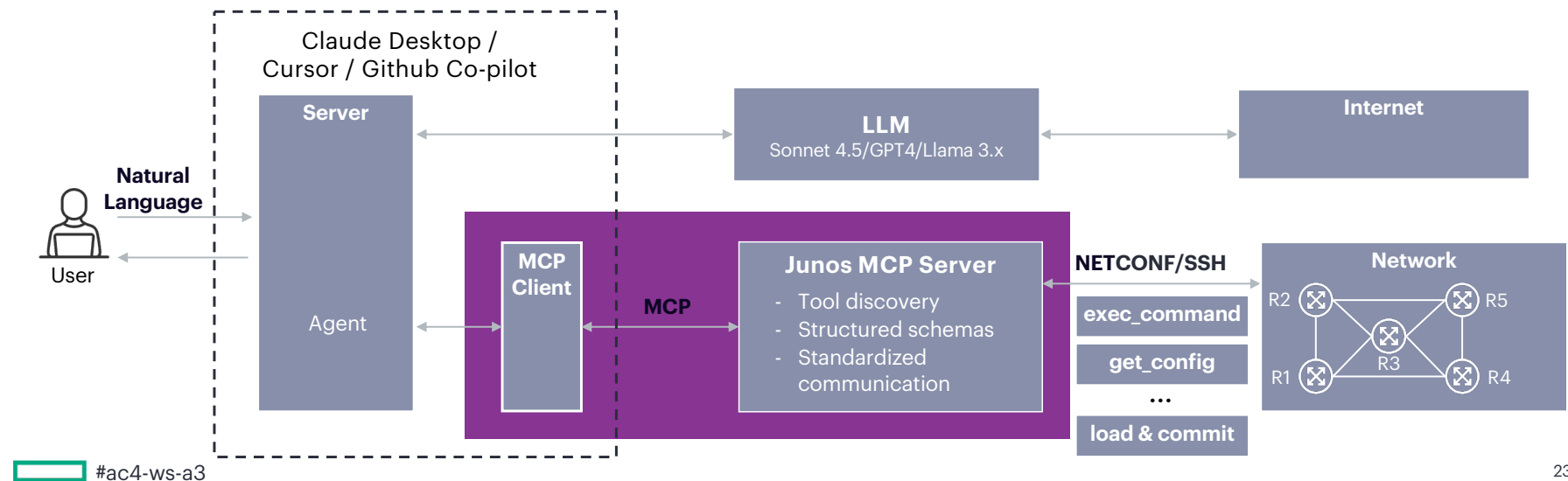
- MCP is a standardized protocol that allows AI models to connect and **interact** with different environments
- Removing the need for custom code speeds up time to market and makes Agentic AI practical
- **Examples:** Search knowledge bases, pull from databases, utilize APIs, gather analytics, run scripts, interact with code repositories, connect to workflow engines...



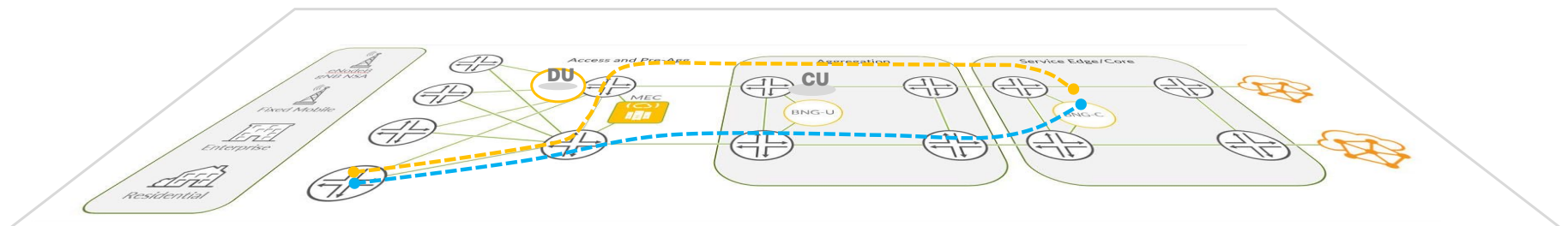
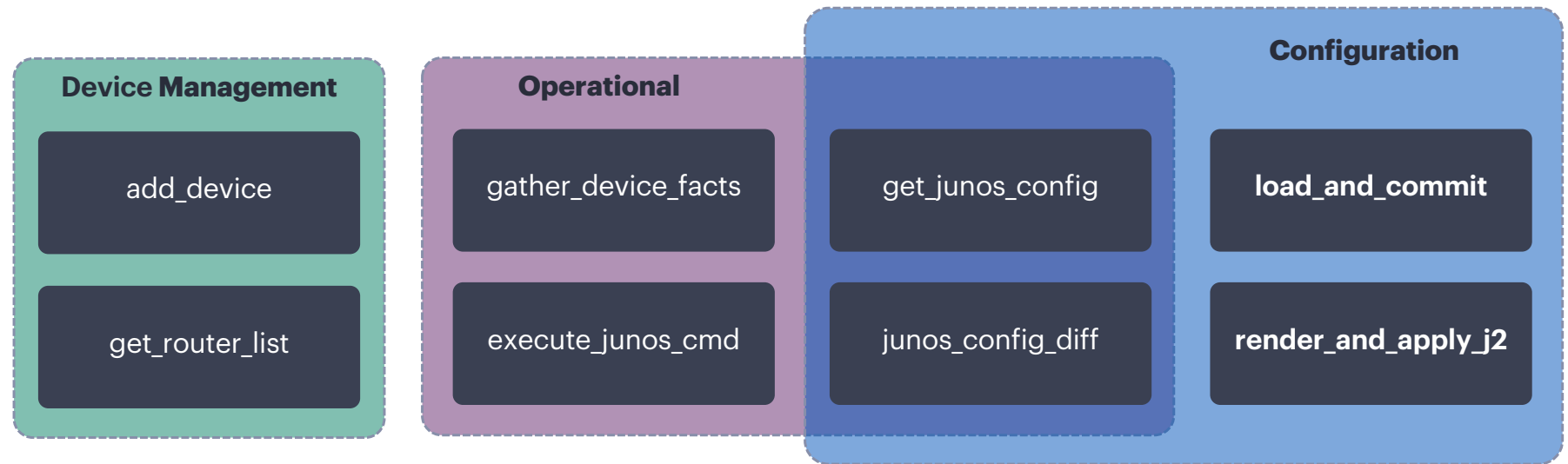
Junos MCP server

MCP server for physical, virtual, and containerized platforms that run Junos

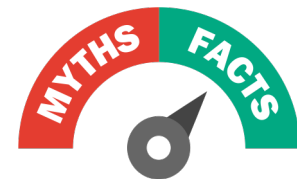
- **Provides tools, schema, and standards** for monitoring, configuration management, and more
- Hosted on Juniper's Github Organizational Account: <https://github.com/Juniper/junos-mcp-server>
- **Open-source**, Apache license 2.0



Junos MCP Server: Tools

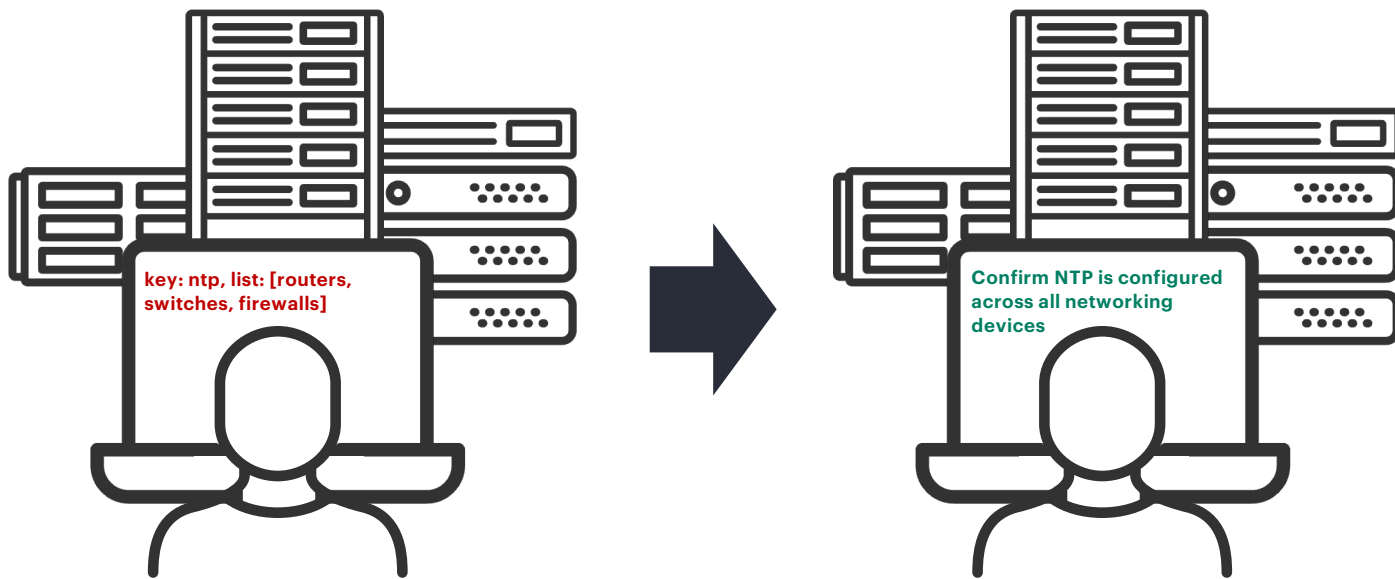


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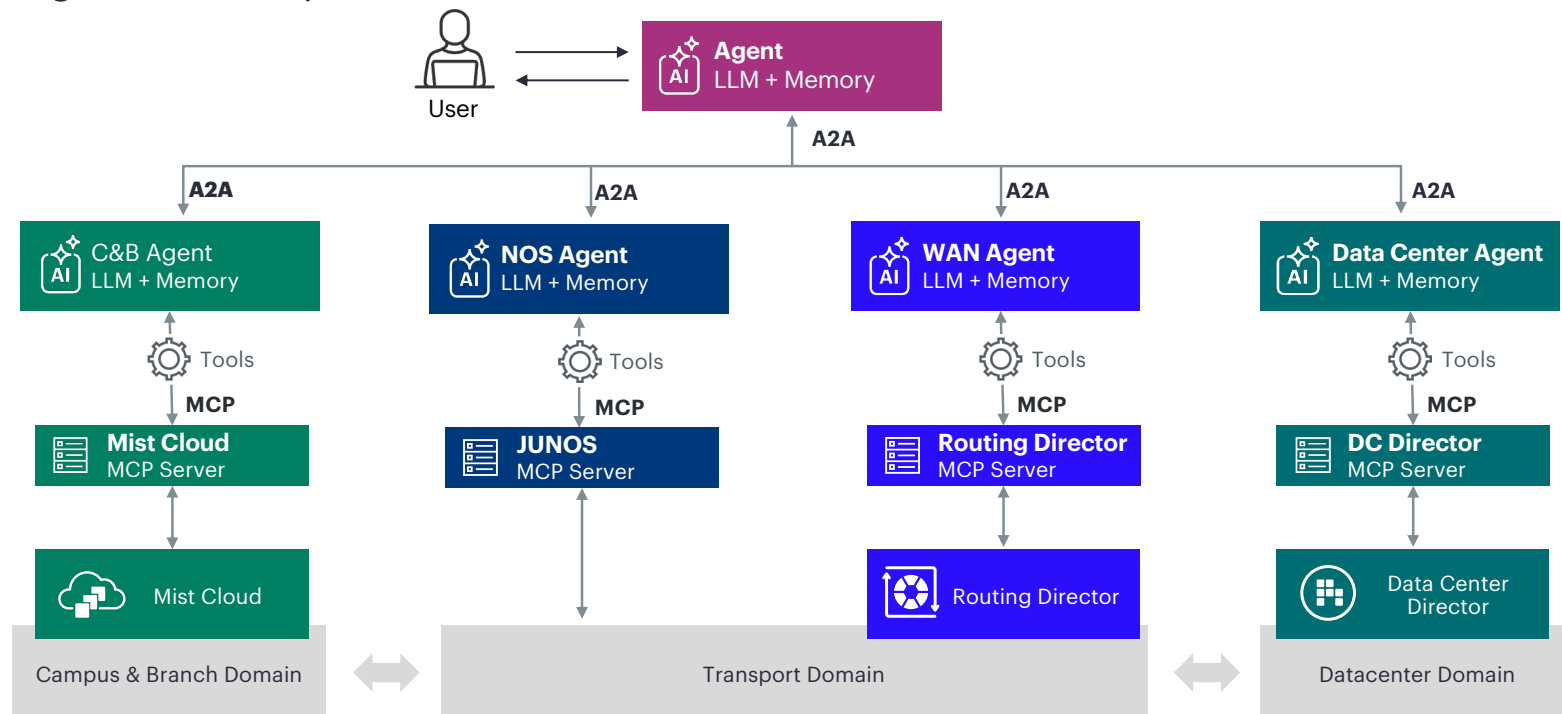
Natural Language

Removes the need to learn how to code or anything do anything code-like



End-to-end Juniper Architecture with A2A and MCP

Agent-to-Agent is an open standard that enables different AI agents to communicate and work together directly



Workshop

Jose Miguel Izquierdo | Nilesh Simaria | Jessica Garrison

 #ac4-ws-a3

Juniper Public



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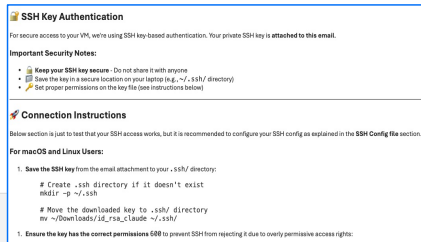


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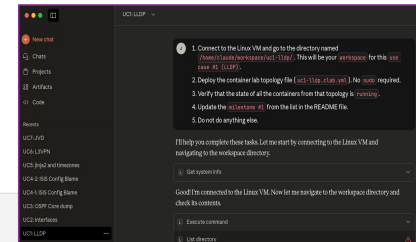
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#ac4-ws-a3

TIPS: Transforming Network Engineers to Automation Practitioners with MCP

BRING LABS UP

```
cd /home/claude/workspace/<Use Case ID> &&  
containerlab deploy -t <Use Case ID>.clab.yml
```

Example:

```
cd /home/claude/workspace/uc1-ldp &&  
containerlab deploy -t uc1-ldp.clab.yml
```

SPLIT TERMINAL

To get a 3x2 view of all 6 devices, "Cmd + d" or "Cmd + Shift + d" to split it either vertically or horizontally.

YOU CAN SEE WHAT CLAUDE IS DOING

If you expand on what the tools are doing, you can see the commands that are being executed on the VM.

BRING LABS DOWN

```
gousecase<Use Case ID>  
clab destroy -c
```

Example:

```
gousecase1  
clab destroy -c
```

SEND COMMANDS TO ALL TERMINALS

"Cmd + Shift + l" to send you command to all 6 terminals

DON'T TEAR DOWN THE LAB, YET

When you complete the lab, the last instruction is to tear it down. If you are following along with us... don't. Play with your topology until we are ready to move to the next lab.

Claude Tips

Claude is slow

- Restart Claude
- Switch your LLM (Sonnet 4 or Sonnet 4.5)

Allow Button

- Pay attention for the first few use cases and be diligent about hitting “Allow”

Token Usage

- Maximizing tokens
- LLM selection
- What not to do

Stop Button

- When you make a mistake or need to stop Claude from completing the goal

Deploy & Teardown

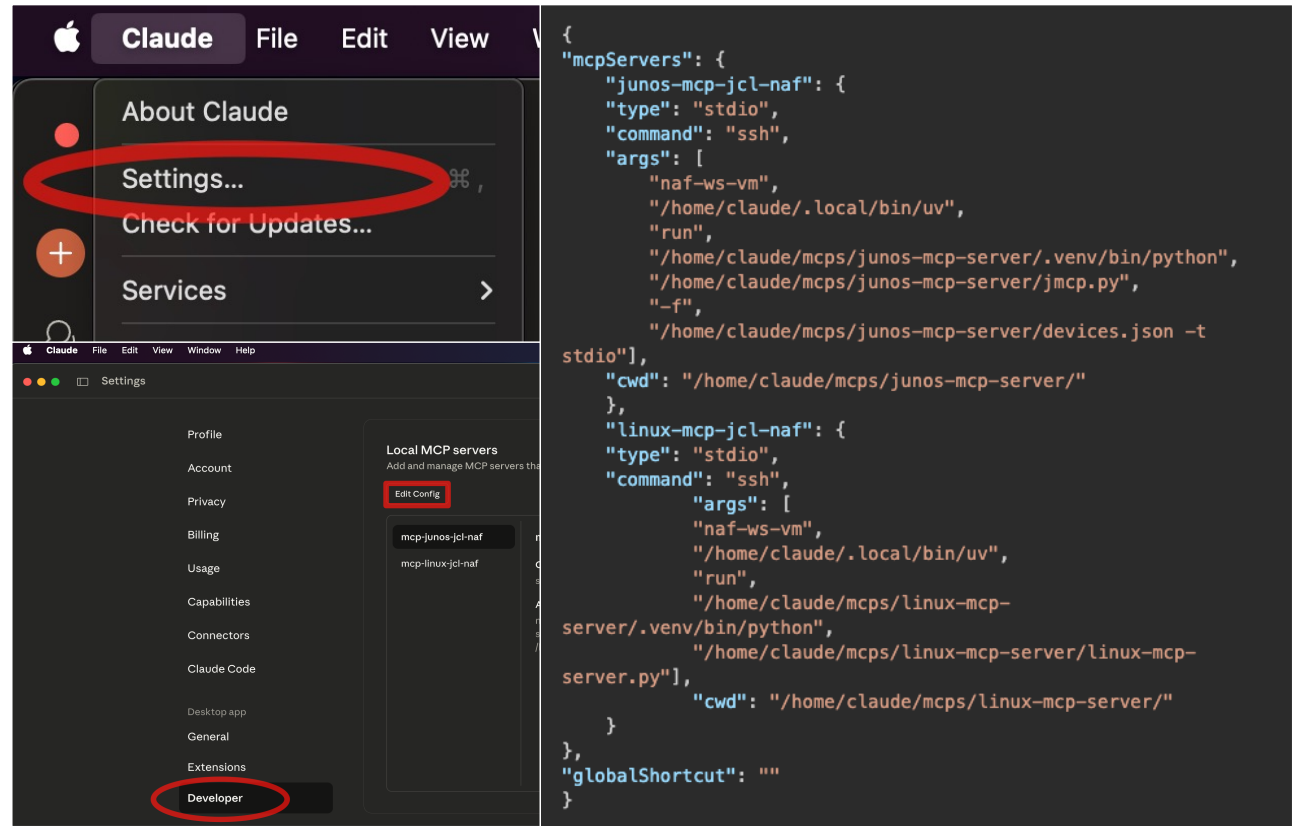
- Can be done manually to save time or tokens

Rename Chat

- Use your own naming schema
- Keep track of use cases

Configure Claude Desktop

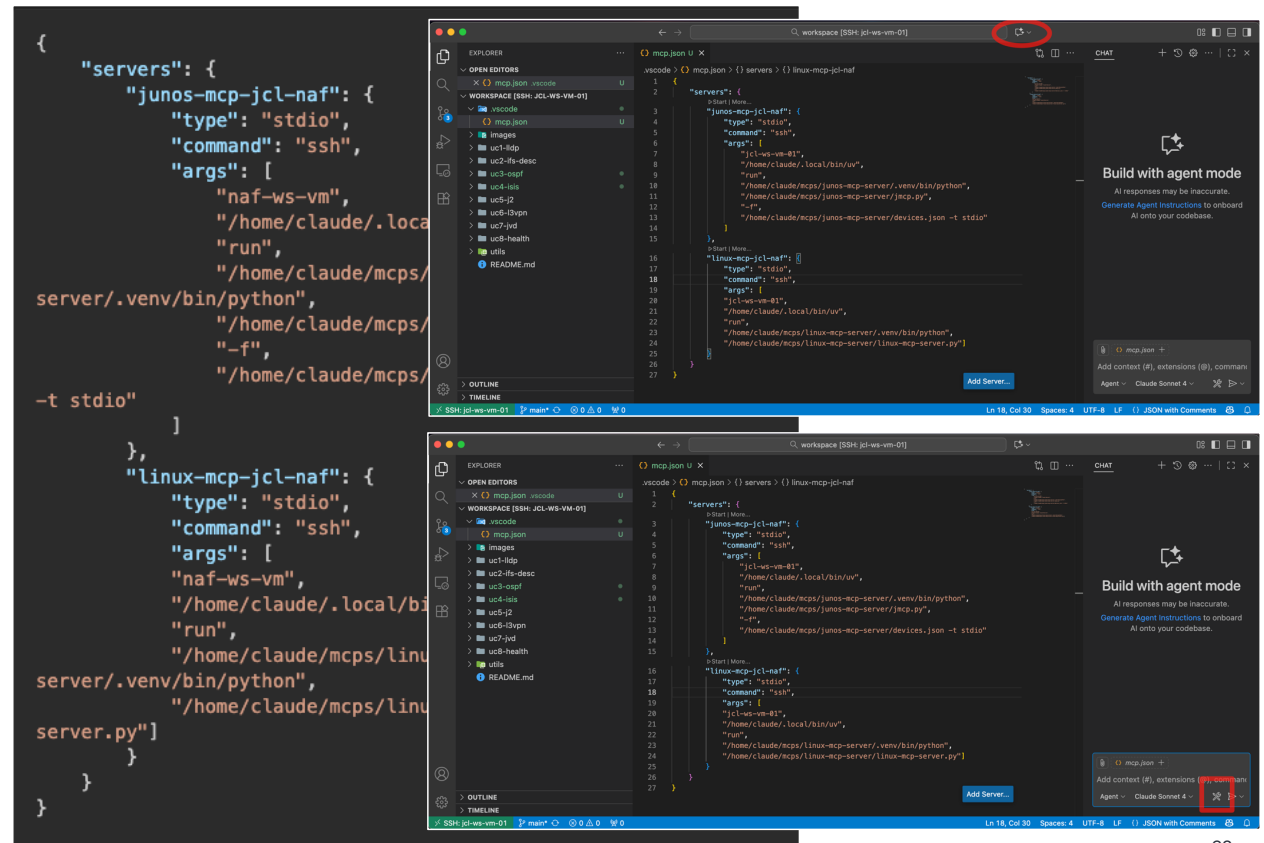
1. Install Claude Desktop
2. Launch Claude Desktop
3. Configure your [license](#); we recommended PRO.
4. Configure Claude Desktop MCP servers
Settings > Developer > Edit Config > Update the JSON
5. Restart Claude
6. Check Claude Desktop logs
7. “Always Allow” button at first usage



Configure GitHub Copilot

Using Visual Studio Code

1. Launch Visual Studio Code
2. Open workspace in Remote VM, add as mcp.json in .vscode folder
3. Open AI Chat
4. Configure Tools in AI Chat
5. Select NAT Workshop MCPs



#ac4-ws-a3

<https://status.claude.com/>

All Systems Operational

Uptime over the past 90 days. [View historical uptime.](#)

claude.ai

Operational



90 days ago 99.71 % uptime Today

platform.claude.com (formerly console.anthropic.com)

Operational



90 days ago 99.84 % uptime Today

Claude API (api.anthropic.com)

Operational



90 days ago 99.81 % uptime Today

Claude Code

Operational

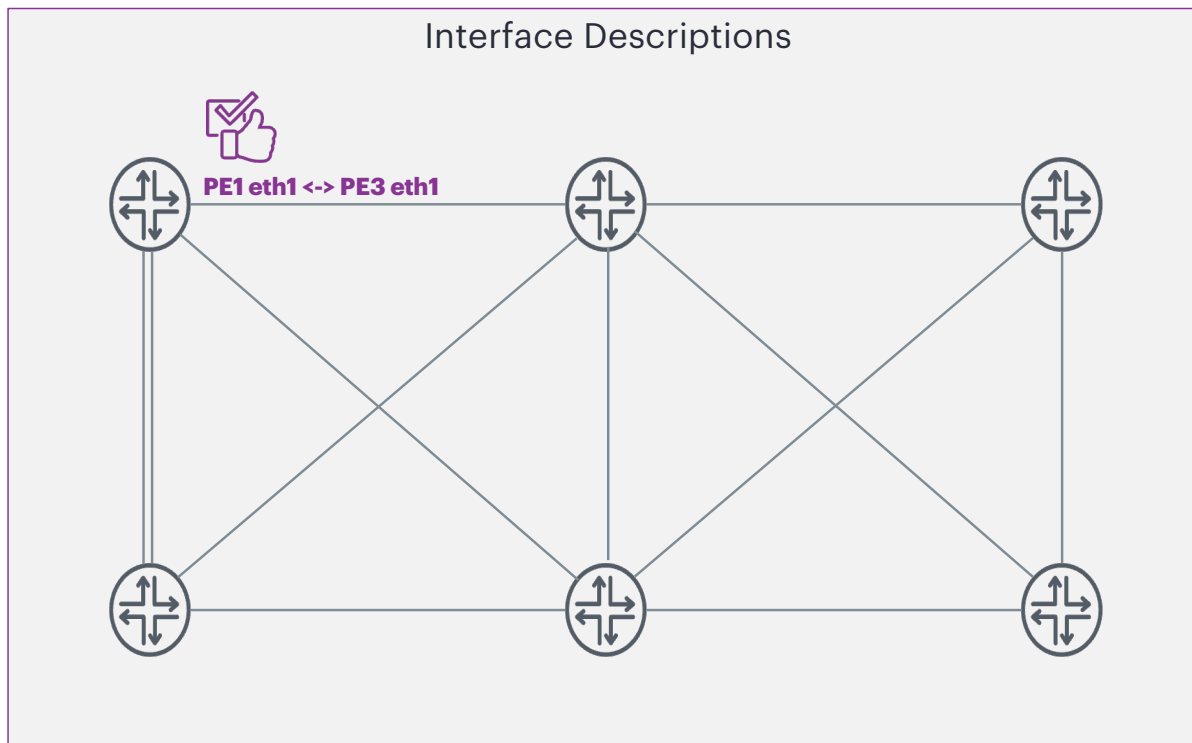


90 days ago 99.79 % uptime Today

Lab 1: Config Audit w/ Interface Descriptions

 Estimated Lab Time:
10:30 - 11am

/home/claude/workspace/uc2-ifs-desc/



AI GOALS:

- Configuration compliance with interface descriptions
- Complies with request for human verification before pushing a configuration change to make the fix
- Created simple report for change

TIPS:

- When you get started with Claude you need to be vigilant with the “Always Allow” button

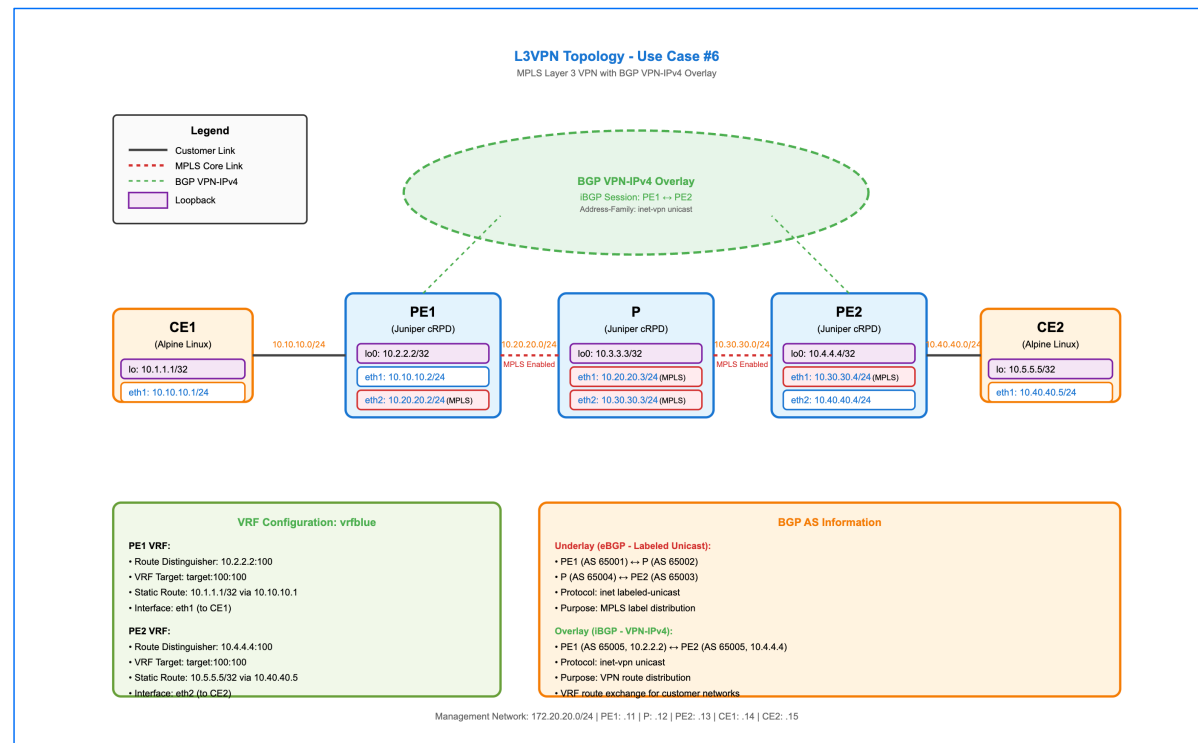
 #ac4-ws-a3

Lab 2: Configuration Support with L3VPN

/home/claude/workspace/uc6-l3vpn/



Estimated Lab Time:
11:15 - 11:45am



AI GOALS:

- Help user configure L3VPN correctly
- Provide root cause analysis, configuration suggestions, and verify results after a config push

FAQs:

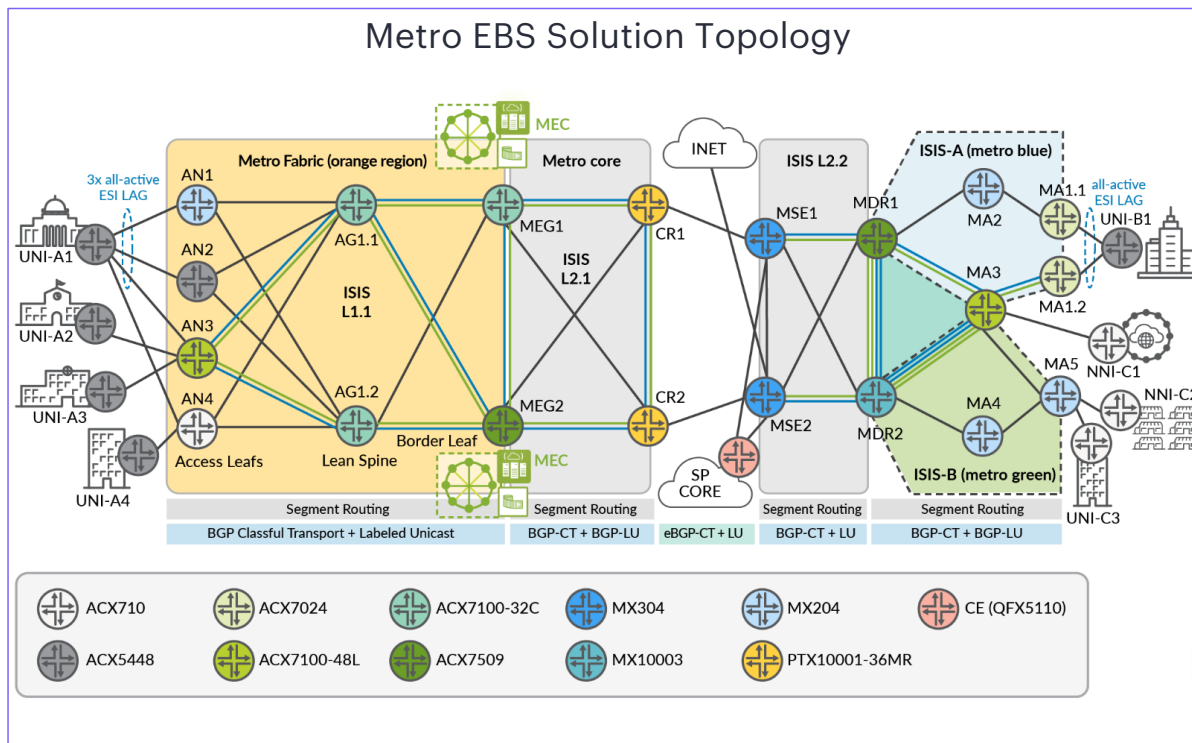
- Incorrect configurations are one of the top reasons customers open cases

#ac4-ws-a3

Lab 3: Juniper Validated Design Deployment

/home/claude/workspace/uc7-jvd/

Estimated Lab Time:
11:45am-12:15pm



AI GOALS:

- Automate the replication of a validated design in a virtual environment

STARTING POINT:

- Can be used to make adjustments to a JVD

LAB IMAGES:

- Our lab uses cRPD, so this is not a digital twin as it lacks a solid data plane. But our free vJunos lab images have data plane emulators.

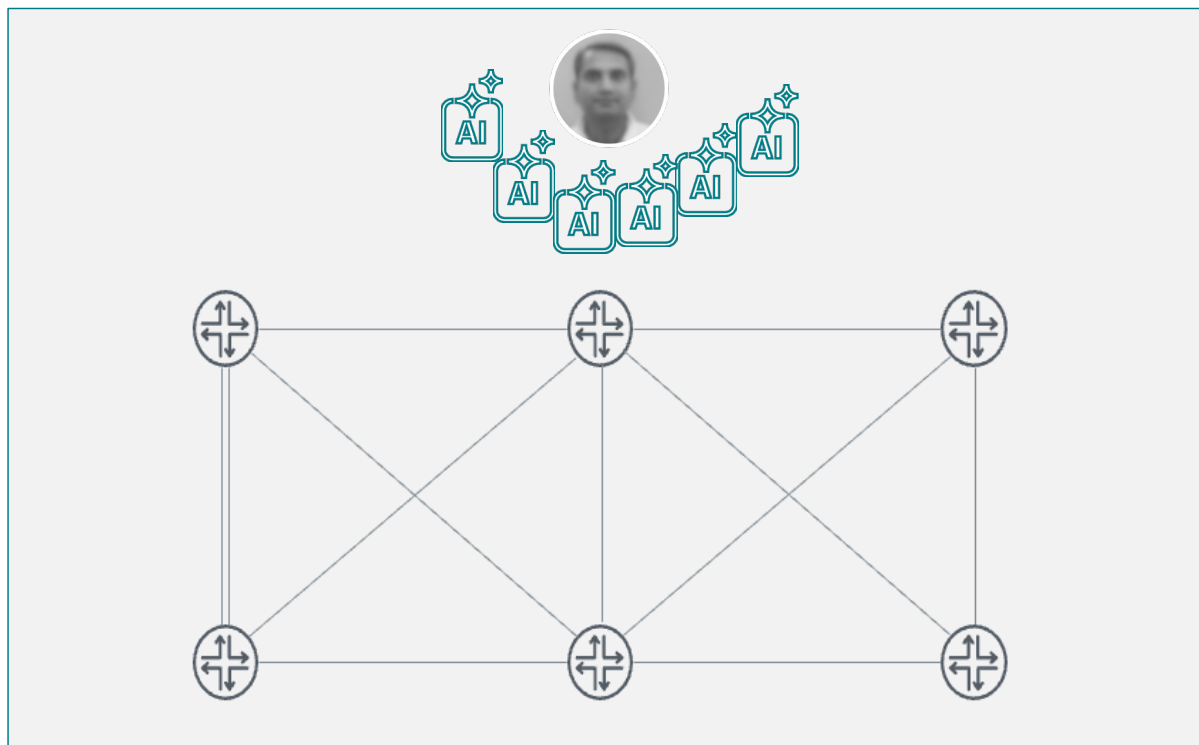
#ac4-ws-a3

Lab 4: Design your own health check

/home/claude/workspace/uc8-health/



Estimated Lab Time:
12:15-1:00pm



AI GOALS:

- Use instructions provided by the user to check device and network health

TIPS:

- You can use any topology from any of the 8 use cases.
- This is the last lab, go for it until you run out of tokens!

#ac4-ws-a3

Thank you

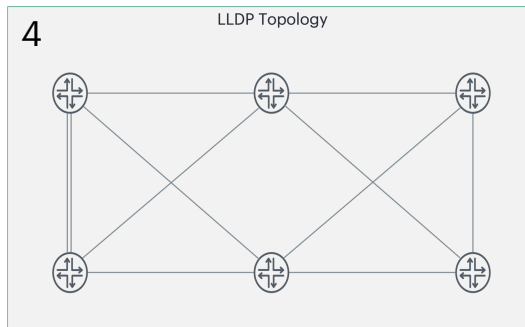
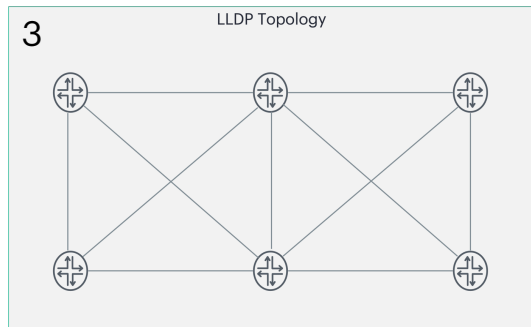
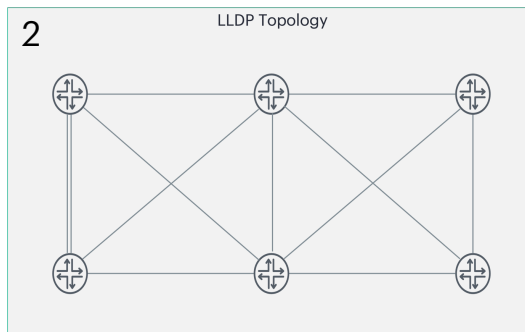
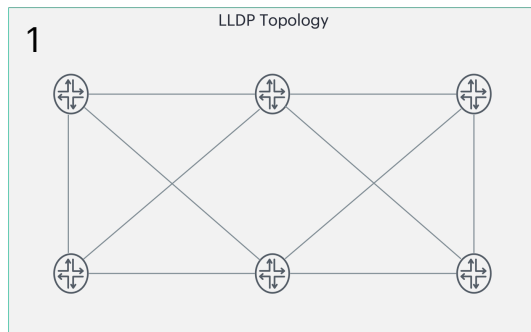
Jose Miguel Izquierdo | Nilesh Simaria | Jessica Garrison

Back of Deck

Eight use cases were built. Our internal BETA workshop attendees voted and chose the top 4 and those are what we are doing today. But here are the others, in case you want to play with them.

Lab 1: LLDP with Topology Diagram

/home/claude/workspace/uc1-lddp/



AI GOALS:

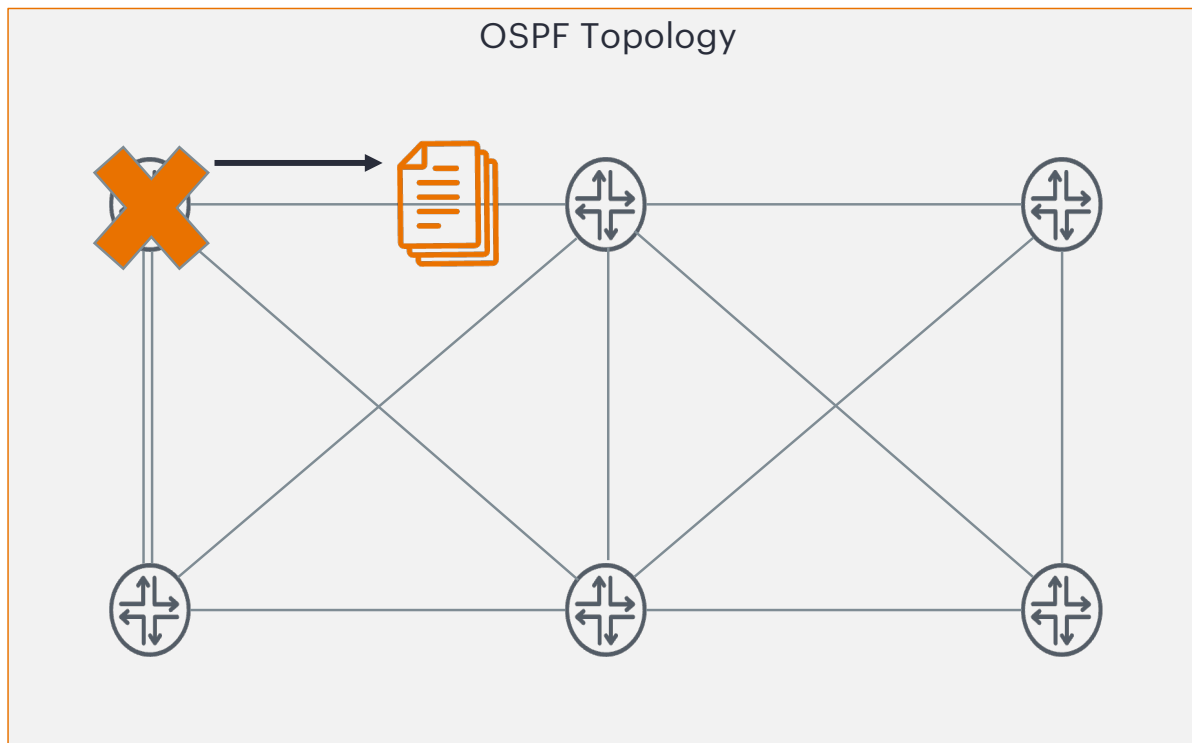
- Autonomous configuration of LLDP, including interface description
- Verification of LLDP configuration, neighbor status,
- Generation of topology diagram

TIPS:

#ac4-ws-a3

Lab 3: Core Dump with OSPF

/home/claude/workspace/uc3-ospf/



AI GOALS:

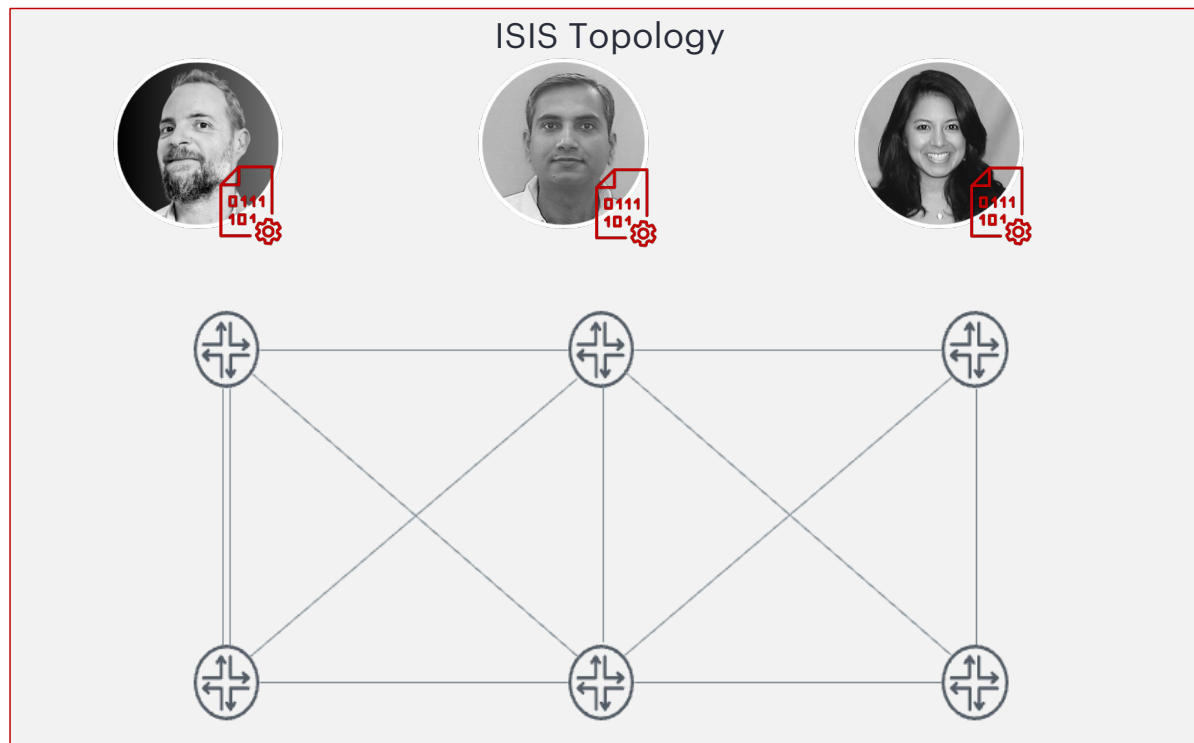
- Core dump analysis and report generation

TIPS:

- Look through the core dump markdown file
- Best practice: Always look at the *show system core-dump* command and do not assume all core dumps will be written to the same location in the directory

Lab 4: Configuration Blame Analysis with ISIS

/home/claude/workspace/uc4-isis/



AI GOALS:

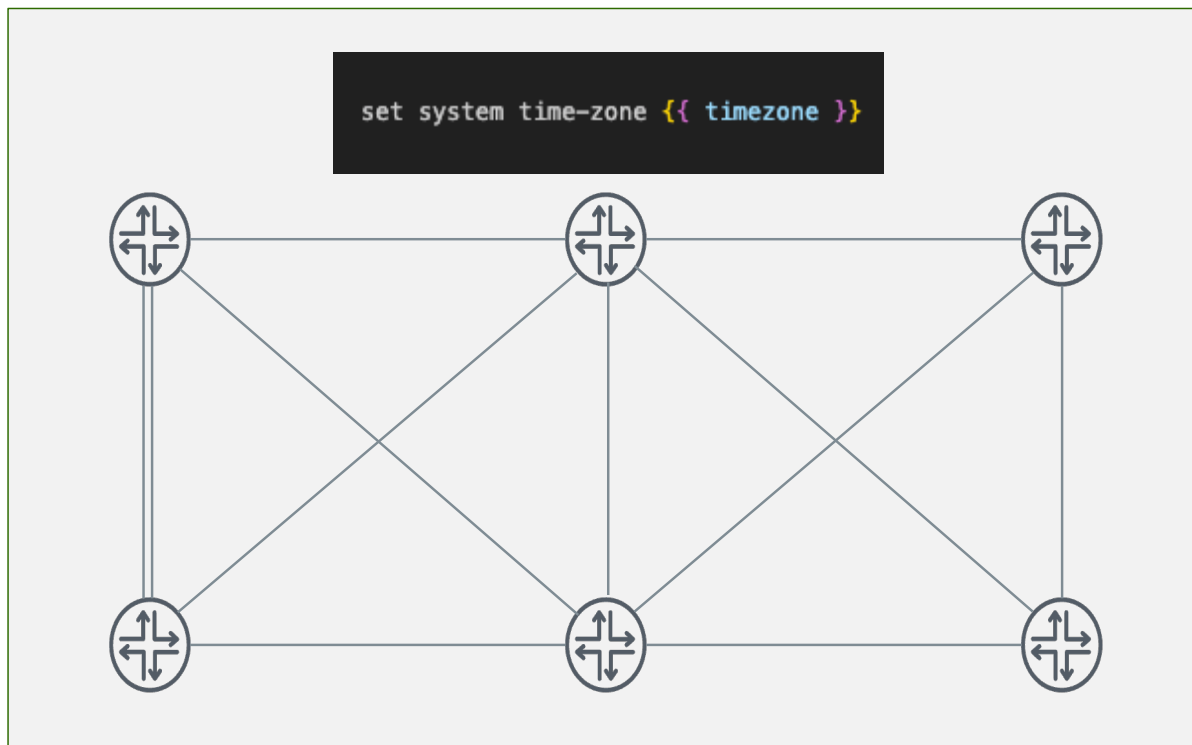
- Utilize the Configuration Blame Analysis Document to find root cause
- Was it Josemi, Niles or Jessica that broke the network?

FAQs:

- This is UC4

Lab 5: Jinja2 Templates

/home/claude/workspace/uc5-j2/



#ac4-ws-a3

Juniper Public

AI GOALS:

- Support user with the tedious task of template creation
- Confirm with one device before pushing to all others

TIPS: